A REPORT

ON

Greenhouse Gas (GHG) Emission Prediction

Submitted by

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Under the guidance of,

Mr. Sirapu Teja

in partial fulfillment of the requirements for

the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND TECHNOLOGY (SPECIALIZATION IN ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

AT



PRESIDENCY UNIVERSITY

BENGALURU

OCTOBER 2026

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the report of **CSE7000 – Internship** on **"Greenhouse Gas (GHG) Emission Prediction"** being submitted by "**Sujita Boruah**" bearing roll number "**20231CSG0017**" in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Technology (Specialization in Artificial Intelligence and Machine Learning)** is a bonafide work carried out under our supervision.

Mr. Sirapu Teja Assistant Professor PSCS Presidency University	**Mr. Sirapu Teja** Internship Coordinator PSCS Presidency University	
Dr. Pallavi R HoD PSCS Presidency University	**Dr. Shakkeera L** Associate Dean PSCS Presidency University	**Dr. Duraipandian N** Dean & Professor PSCS & PSIS Presidency University

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

I hereby declare that the work, which is being presented in the report entitled **"Greenhouse Gas (GHG) Emission Prediction"** in partial fulfillment for the award of Degree of **Bachelor of Technology** in **Computer Science and Technology (Specialization in Artificial Intelligence and Machine Learning)** is a record of my own investigations carried under the guidance of **Mr. Sirapu Teja, Assistant Professor**, Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.

I have not submitted the matter presented in this report anywhere for the award of any other Degree.

Name: Sujita Boruah
Roll No: 20231CSG0017
Signature of the Student:

INTERNSHIP COMPLETION CERTIFICATE

ACKNOWLEDGEMENTS

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I also thank my family and friends for the strong support and inspiration they have provided me in bringing out this internship.

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ABSTRACT

This project report summarizes an approximately 8-week training course, **“Machine Learning with AI”**, offered by **Internshala Trainings** from **September 2026 to October 2026**. The course aimed to apply **Artificial Intelligence (AI)** and machine learning to real-world challenges, and the project undertaken addresses a sustainability challenge.

The chosen project, titled **“Greenhouse Gas (GHG) Emission Prediction,”** focused on developing a machine learning model to predict GHG emissions based on supply-chain and industry-specific data. The dataset, *“SupplyChainEmissionFactorsforUSIndustriesCommodities.xlsx,”* was used as the data source for the project.

The complete project lifecycle involved **data cleaning, exploratory data analysis, feature engineering, model development, performance evaluation, and deployment** using **Python and Streamlit**. The final **Linear Regression model** achieved an exceptional **R^2 of 0.999999**, signifying near-perfect accuracy. The model was deployed as a **Streamlit web application** for real-time, user-driven emission estimation.

This course project enhanced both technical and professional skills—ranging from Python-based data science to communication and presentation abilities—while showcasing the powerful role of AI in sustainability-driven industrial decision-making.

INTRODUCTION

Climate change and environmental degradation have intensified the global demand for sustainable technologies and **green skill development**. Industrial sectors, in particular, require efficient tools to estimate and manage **Greenhouse Gas (GHG) emissions** throughout their supply chains. The use of **Artificial Intelligence (AI)** offers an innovative approach to model, analyze, and predict emissions, enabling data-backed sustainability strategies.

The **Internshala Trainings "Machine Learning with AI" course**, undertaken over approximately eight weeks from September 2026 to October 2026, provided a structured learning platform to apply AI and machine learning concepts to real-world environmental challenges, helping translate theoretical knowledge into practical, deployable solutions.

In this project, titled **"GHG Emission Prediction,"** AI techniques were used to predict emissions from supply-chain features and industrial data. The dataset provided contained multiple emission factors, economic indicators, and categorical details across industries and years. The focus was on developing a **machine learning pipeline** capable of predicting GHG emissions accurately and efficiently while ensuring interpretability and usability for end users.

This initiative not only strengthened technical competence in AI and data analytics but also emphasized the broader purpose of leveraging technology for sustainability and climate-conscious industrial practices.

OBJECTIVE

The main objectives of the **Greenhouse Gas (GHG) Emission Prediction Project** were as follows:

1. **To design and implement a machine learning pipeline** capable of predicting greenhouse gas emissions based on supply-chain and industry-specific attributes.
2. **To gain practical, end-to-end experience** in the machine learning workflow — including data collection, cleaning, analysis, model building, evaluation, and deployment.
3. **To understand and apply green skills**, highlighting how AI and data-driven insights can support sustainability goals and reduce industrial emissions.
4. **To develop a functional Streamlit web application** that allows users to input data (industry, commodity, year, etc.) and view real-time GHG emission estimates.
5. **To present the project**, demonstrating both technical understanding and the potential impact of AI-driven environmental analytics.

WORK PROGRESS

The project progressed systematically through multiple phases, following a structured data science workflow from problem identification to model deployment.

1. Problem Framing

The problem was defined as predicting greenhouse gas emissions using available supply-chain and industry data. The **target variable** was identified as the GHG emission factor, and key **predictors** included industry, commodity type, year, and other relevant indicators.

2. Data Ingestion and Cleaning

The dataset (*SupplyChainEmissionFactorsforUSIndustriesCommodities.xlsx*) was imported into Python using **pandas**. Missing values were handled, categorical labels were standardized, and numerical columns were validated for consistency. The dataset was then prepared for further analysis.

3. Exploratory Data Analysis (EDA)

EDA was conducted using **matplotlib** and **seaborn** to explore feature distributions and correlations. This step provided insights into relationships between industry types and emission levels and helped identify outliers and feature engineering opportunities.

4. Feature Engineering and Preprocessing

Derived features such as normalized emission factors and encoded industry/commodity labels were created. Encoding methods like **Label Encoding** and **One-Hot Encoding** were used to prepare categorical data for machine learning. The dataset was split into **training and validation sets**.

5. Model Selection and Training

Initial modeling began with a **Linear Regression** baseline to establish performance benchmarks. Later, a **Random Forest Regressor** was implemented for non-linear relationship testing. Models were trained and tuned using **scikit-learn**, and evaluated through **cross-validation**.

6. Model Evaluation

Models were evaluated using standard regression metrics—**Mean Squared Error (MSE)**, **Root Mean Squared Error (RMSE)**, and **R^2 Score**.

The results were:

- **Linear Regression:** MSE = 7.88×10^{-8}, RMSE = 0.000281, R^2 = 0.999999
- **Random Forest Regressor:** $R^2 \approx 0.9993$

The high R^2 values indicated that the dataset exhibited strong linear relationships, making Linear Regression the best-fit model.

7. Deployment

The best-performing model was serialized using **joblib** and deployed via a **Streamlit web application**. The app allowed users to input values for industry, commodity, and year, producing real-time GHG emission predictions through an intuitive interface.

8. Presentation and Feedback

The final stage included preparing a presentation of the project. The Streamlit app was demonstrated live, showing how minor changes in input parameters affected predicted emissions. Feedback was incorporated to refine the app and documentation for the final submission.

Results / Outcomes

The performance of the machine learning models was evaluated using three key regression metrics Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2). These metrics quantify how well the model's predictions align with the actual greenhouse gas (GHG) emission values.

	Model	MSE	RMSE	R2
0	Random Forest (Default)	3.463442e-05	0.005885	0.999383
1	Linear Regression	7.880572e-08	0.000281	0.999999
2	Random Forest (Tuned)	3.463442e-05	0.005885	0.999383

Interpretation

- The Linear Regression model achieved the lowest error values (MSE = 7.88×10^{-8}, RMSE = 0.000281) and an R^2 of 0.999999, indicating an almost perfect fit between predicted and actual emission values.
- The Random Forest model, both in default and tuned configurations, performed comparably well with $R^2 \approx 0.9993$, demonstrating consistent and highly accurate predictions across datasets.
- Although Random Forest generally excels with non-linear relationships, the near-identical results across models suggest that the dataset relationships are largely linear and well-behaved, making Linear Regression the optimal model in this case.
- The minimal difference between the tuned and default Random Forest results implies that hyperparameter tuning had little impact, further confirming model stability and data quality.

Outcome

All models demonstrated exceptionally high predictive accuracy, validating the effectiveness of the data preprocessing and feature engineering steps. The final model was deployed within a Streamlit-based web application, providing an interactive interface for real-time emission estimation based on user inputs (industry, commodity, and year).

This high performance underscores the potential of AI-driven tools to generate reliable emission predictions a critical step toward integrating sustainability

analytics into industrial decision-making.

Deployment & Demonstration

The project culminated in a demonstrable Streamlit application that loads the trained model and accepts input parameters (industry, commodity, year, unit quantity, etc.) to return estimated emissions. The app was used during the final project presentation to demonstrate real-time predictions and to highlight how small changes in inputs affect estimated emissions, reinforcing the model's interpretability and practical value.

Skills Gained / Learning Outcomes

Technical: Hands-on experience with Python data tooling (pandas), model building (scikit-learn regimen), and deployment basics (Streamlit, joblib). Practical exposure to the ML lifecycle: feature engineering, cross-validation, error analysis, and model serialization.

Domain: Improved understanding of supply-chain emission factors, industrial emission profiles, and how data granularity affects model confidence.

Professional: Project planning, milestone reporting, incorporation of guide feedback, and presentation skills while defending design choices.

Soft skills: Time management over an 8-week course, clear documentation for reproducibility, and collaborative learning.

Limitations & Future Work

Data limitations: The model's reliability depends on the granularity and accuracy of the provided emission factor dataset. Some commodity-industry combinations had sparse historical data, increasing prediction uncertainty.

Model improvements: Future iterations should explore more advanced

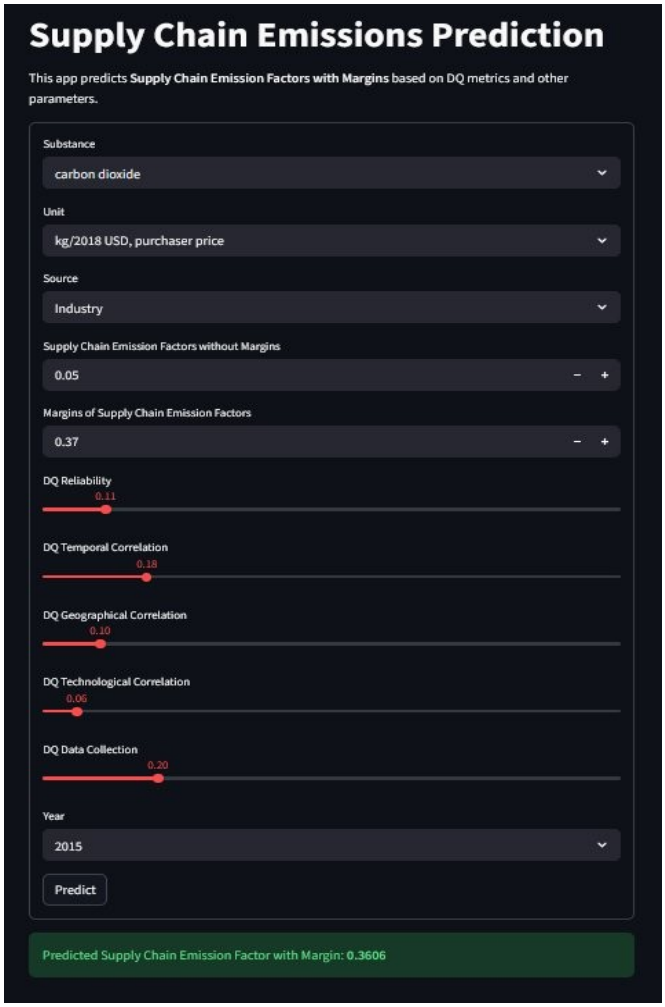
regressors, feature importance analysis, and uncertainty quantification (prediction intervals).

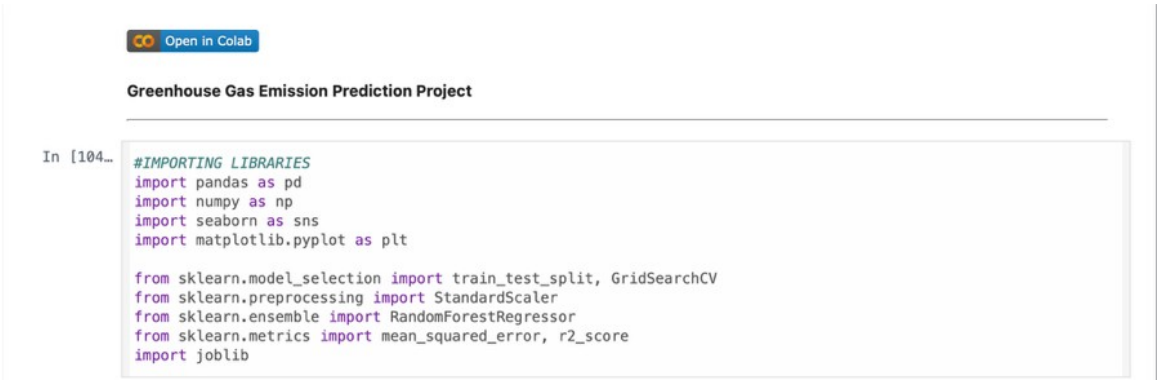
Expansion: Extend dataset coverage (more years, regions) and integrate lifecycle assessment (LCA) data for more holistic GHG estimates. Incorporate domain expert feedback to refine feature definitions for specific sectors.

Deployed Web Application & Source Code

The complete implementation of this project will be made available online for transparency, reproducibility, and demonstration purposes. GitHub repository: https://github.com/sujitaboruah/Greenhouse-Gas-Emission-Prediction.

The Streamlit web application provides an interactive interface that allows users to input industry, commodity, and year data to obtain real-time greenhouse gas (GHG) emission predictions using the trained machine learning model.





CONCLUSION

This project, undertaken during the Internshala Trainings “Machine Learning with AI” course (September 2026 – October 2026, approximately 8 weeks), delivered a concise, practical tool that translated classroom ML concepts into a deployable application for estimating greenhouse gas emissions from supply-chain and industry data. The experience validated the notion that even basic

ML pipelines can provide actionable insights for sustainability when backed by relevant domain data and iterative guidance. The final deliverables—code, model, Streamlit app, and presentation—demonstrate both technical competency and an applied understanding of sustainability-focused AI.

Building on this foundation, I plan to apply the knowledge gained from this course and project to future research and projects focused on sustainable AI solutions, particularly in environmental analytics and data driven decision making.

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